Worked solutions to selected exercises from
G. Chrystal, *Algebra: An Elementary Text-Book*, Part I (1886)

1 The Fundamental Laws and Processes of Algebra

Exercises I

1. Point out in what sense the usual arrangement of the multiplication of 365 by 492 is an instance of the law of distribution.

2. I have a multiplying machine, but the most it can do at one time is to multiply a number of 10 digits by another number of 10 digits. Explain how I can use my machine to multiply 13693456783231 by 46381239245932.

3. To divide 5004 by 12 is the same as to divide 5004 by 3, and then divide the quotient thus obtained by 4. Of what law of algebra is this an instance?

4. If the remainder on dividing N by a be R, and the quotient P, and if we divide P by b and find a remainder S, show that the remainder on dividing N by ab will be $aS + R$. Illustrate with $5015 \div 12$.

5. Show how to multiply two numbers of 10 digits each so as to obtain merely the number of digits in the product, and the first three digits on the left of the product. Illustrate by finding the number of digits, and the first three left-hand digits in the following:

a) $3659893456789325678 \times 342973489379265$;

b) 2^{64}.

6. Express in the simplest form—

$$-(-(-(-(\ldots (-1) \ldots)))),$$

a) where there are $2n$ brackets;

b) where there are $2n+1$ brackets; n being any whole number whatever.

7. Simplify and condense as much as possible—

$$\boldsymbol{2a - \{3a - [a - (b - a)]\}.}$$

8. Simplify—

a) $3\{4 - 5[6 - 7(8 - 9 \cdot \overline{10 - 11})]\}$;

b) $\frac{1}{3}\{\frac{1}{4} - \frac{1}{5}[\frac{1}{6} - \frac{1}{7}(\frac{1}{8} - \frac{1}{9} \cdot \overline{\frac{1}{10} - \frac{1}{11}})]\}$.

9. Simplify—

$$\boldsymbol{1 - (2 - (3 - (4 - \ldots (9 - (10 - 11)) \ \ldots \ ))).}$$

10. Distribute the following products:

a) $(a+b) \times (a+b)$;

b) $(a-b) \times (a+b)$;

c) $(3a-6b) \times (3a+6b)$;

d) $(\frac{1}{3}a - \frac{1}{6}b) \times (\frac{1}{3}a + \frac{1}{6}b)$.

11. Simplify, by expanding and condensing as much as possible—

$$\boldsymbol{\begin{gathered}\{(m+1)a + (n+1)b\}\{(m-1)a + (n-1)b\} \\ + \{(m+1)a - (n+1)b\}\{(m-1)a - (n-1)b\}.\end{gathered}}$$

12. Simplify—

$$\boldsymbol{\left(x + \frac{1}{x}\right)\left(y + \frac{1}{y}\right) + \left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right).}$$

13. Simplify—

$$\boldsymbol{\left(x + \frac{1}{x}\right)\left(y + \frac{1}{y}\right)\left(z + \frac{1}{z}\right) - \left(x - \frac{1}{x}\right)\left(y - \frac{1}{y}\right)\left(z - \frac{1}{z}\right).}$$

14. Expand and condense as much as possible—

$$\boldsymbol{(\tfrac{1}{2}x + \tfrac{1}{4}y + \tfrac{1}{8}z)(\tfrac{1}{2}x - \tfrac{1}{4}y + \tfrac{1}{8}z).}$$